

Yu Hon, ON (Henry)

London | (+44) 75683 35858 | henryon1130@gmail.com | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | No sponsorship required

SUMMARY

Data Scientist with a PhD from Imperial College London, specialising in healthcare AI and computer vision. Experienced in building end-to-end ML pipelines for imaging and clinical prediction, from data curation and preprocessing to model training, validation, and evaluation on HPC systems.

KEY SKILLS

- **Machine Learning & Data Science:** Python, PyTorch, PyTorch Lightning, scikit-learn, MONAI, TorchIO, NumPy, pandas, feature engineering, model evaluation, calibration, representation learning, transformers
- **ML Engineering:** Modular codebases, configuration-driven pipelines, batch inference, experiment tracking (TensorBoard), reproducible workflows, HPC / multi-GPU training, Git, Docker
- **Domain Expertise:** Medical image analysis, cardiac MRI, oncology imaging, radiotherapy planning workflows
- **Languages:** Cantonese (native), Mandarin & English (professional)

PROFESSIONAL EXPERIENCE

PhD Researcher - Interpretable Cardiac MRI AI

London

National Heart & Lung Institute, Imperial College London

Oct 2021 - Dec 2025

- Built ML/AI models for atrial/fibrosis segmentation from LGE MRI and automated classification of valvular heart disease (aortic stenosis, aortic/mitral regurgitation) from cine MRI.
- Applied post-hoc interpretability methods (saliency mapping, Grad-CAM) to make model predictions explainable to clinical stakeholders.
- Leveraged self-supervised representation learning on unlabelled cardiac MRI to improve performance in annotation-scarce settings, validated across multi-centre datasets and public benchmarks.

Research Assistant & Graduate Teaching Assistant

London

Imperial College London

Apr 2023 - Mar 2024

- Built radiomics feature extraction pipelines for high-throughput microscopy; benchmarked classical ML models, reducing manual analysis effort by 80%.
- Supported 100+ students in computer vision and image processing practicals.

Research Assistant

Guangzhou, China

Sun Yat-sen University Cancer Centre

Oct 2019 - Oct 2021

- Developed CNN and statistical prediction models using multimodal omics and 3D imaging data from 180+ patient cases; supported 500+ dosimetry planning and radiotherapy QA in a regulated clinical environment.

Geophysicist

Crawley

Viridien

Mar 2018 - Mar 2019

- Applied large-scale signal processing to complex volumetric imaging datasets; consolidated two-tool visualisation workflow into one, reducing complexity by 50%.

EDUCATION

Imperial College London

London

PhD in Clinical Medicine Research

2021 - 2025

Imperial College London

London

MSc in Biomedical Engineering

2016 - 2017

University of Essex

Colchester

BEng (Hons) in Electronic Engineering

2012 - 2016